

## CSCE 628 COMPUTATIONAL BIOLOGY · MODULE 06 · LECTURE NOTES

# Multiple Alignment and Phylogeny

*Where error compounds, and the tree looks more certain than it is.*

## MODULE OUTCOMES

- Explain why exact multiple alignment is intractable.
- Explain progressive alignment and its error mode.
- Explain distance and character-based tree methods.
- Explain bootstrap support and what it measures.
- Build and critique a phylogeny.

## 1 Multiple alignment

### EXACT MULTIPLE ALIGNMENT IS EXPONENTIAL IN THE NUMBER OF SEQUENCES, SO EVERY TOOL IS A HEURISTIC

The dynamic programming of Module 02 generalises directly to  $k$  sequences as a  $k$ -dimensional table —  $O(n^k)$  time and space — which is infeasible beyond three or four sequences of any realistic length.

And the problem is NP-hard under the usual sum-of-pairs scoring schemes, so no exact method is coming (CSCE 629 Module 12's framing: this is a settled question, not an open one).

So progressive alignment is what every tool does: align the most similar pair first, then add the remaining sequences one at a time in the order given by a guide tree — which is fast and is unavoidably greedy.

And greedy means early errors are permanent — 'once a gap, always a gap', because a gap introduced in an early pairwise step is carried through every subsequent addition — which is the characteristic failure of every progressive aligner and is precisely why iterative refinement methods exist (§2).

## 2 The circularity

**THE GUIDE TREE NEEDS AN ALIGNMENT AND THE ALIGNMENT NEEDS A GUIDE TREE**

**Progressive alignment orders the sequences by a guide tree, which is built from pairwise distances — which are themselves derived from pairwise alignment scores (Module 03 §1).**

**So the multiple alignment depends on a rough tree, and the final published tree is built from that alignment — which means the tree is not independent evidence for the alignment, and the alignment is not independent evidence for the tree.**

**And iterative methods alternate between the two until the result is stable — which converges to something, and not necessarily to the right thing (CSCE 669 Module 04's local optima: alternating optimisation has no global guarantee).**

**Which is a good example of a dependency that is easy to miss when reading a figure — the alignment and the tree presented in the same paper are not two independent results, though they are frequently read as mutually corroborating.**

### 3 Building trees

| Method             | How it works                                                                          | What it assumes                                                                                              |
|--------------------|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| UPGMA              | Hierarchical clustering by average distance.                                          | A molecular clock — equal rates on all lineages, which is usually false and produces characteristic errors.  |
| Neighbour-joining  | Greedily minimise total branch length.                                                | Additive distances. Fast, widely used, and reasonable when divergence is moderate.                           |
| Maximum parsimony  | The tree requiring the fewest character changes.                                      | That changes are rare. Susceptible to long-branch attraction — see §4.                                       |
| Maximum likelihood | The tree maximising the probability of the data under an explicit evolutionary model. | The model. Slow, and principled — and the model choice is itself reportable.                                 |
| Bayesian inference | Sample the posterior distribution over trees.                                         | The model and the prior. Gives credible intervals rather than a point estimate, which is its main advantage. |

Every method assumes something that is not exactly true — and the useful question is not 'is the assumption true' but 'would its failure produce the tree I am looking at', which is answerable and specific. The assumption column is the one to read before believing a tree, and a paper that does not name its model has not told you enough to apply the question.

### 4 Support

- The bootstrap resamples the alignment's columns with replacement and rebuilds the tree from each replicate — the support value on a branch is the fraction of replicates in which that branch appears (CSCE 658 Module 08's bootstrap, applied to trees).
- Which measures whether the *alignment* supports the branch consistently across its columns — not whether the branch is true — and the two come apart whenever the evolutionary model is wrong in a way that affects all columns similarly.
- So systematic error produces high support for the wrong tree — long-branch attraction being the classic case, where rapidly evolving lineages are grouped because of shared convergent changes, and more data increases confidence in the error rather than correcting it.
- And the alignment's own uncertainty is excluded entirely (§1) — the bootstrap resamples columns of a single fixed alignment, treating that alignment as given when it is itself a heuristic output with its own error.
- Which means published support values are optimistic, structurally, and should be read as a lower bound on the uncertainty rather than as a calibrated probability. The bootstrap treats the alignment as given, so it excludes the largest source of uncertainty in the whole procedure.

**AND A TREE IS A HYPOTHESIS DRAWN WITH GREAT CONFIDENCE**

A phylogeny is rendered as a clean branching diagram with crisp bifurcations — which is a visual claim of certainty that the underlying inference does not support (CSCE 679 Module 09 §1: a point estimate drawn without its uncertainty invites a conclusion the data does not support, here in a discrete setting).

And the errors compound along the pipeline: alignment error feeds tree error, and the tree is then used to interpret the alignment and sometimes to refine it — which is §2's circularity with consequences.

So the honest presentation shows support values on every branch, states the evolutionary model and the software version, and collapses poorly supported branches into polytomies — which is routine good practice and is not universal.

Which is this module's version of the program's rule: draw the uncertainty, name the model, and do not let a tidy diagram do the claiming for you — because a reader looking at a tree sees a fact, not an inference.

## Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

| Resource                                                                              | What it covers for this module                                                 |
|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Durbin et al., chapters 6 and 7 (library copy)                                        | §§1 and 3 — multiple alignment and phylogeny, probabilistically.               |
| Felsenstein — Inferring Phylogenies                                                   | §§3 and 4 — the standard text, and the bootstrap is his. Library copy.         |
| Edgar & Batzoglou — Multiple sequence alignment (free review)                         | §§1 and 2 — the heuristics compared, with their failure modes.                 |
| Felsenstein — Cases in which parsimony or compatibility will be positively misleading | §4's long-branch attraction, in the original — and the title says the finding. |

## Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Count the cells** in a  $k$ -dimensional alignment table for  $k = 5$ ,  $n = 300$ .
- 2 **Implement progressive alignment** for four short sequences.
- 3 **Change the guide tree order** and compare the alignments.
- 4 **Construct a case** where an early gap is clearly wrong and cannot be fixed.
- 5 **Build trees by two methods** from the same alignment and compare.

- 6 **Build a tree from two different alignments** of the same sequences.
- 7 **Run a bootstrap** with 1000 replicates and report support.
- 8 **Simulate long-branch attraction** and watch support rise for the wrong grouping.
- 9 **Collapse branches below 70% support** and compare the two figures.
- 10 **Find a published tree** with no support values, and say what is missing.

## Self-check

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Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Why is exact multiple alignment infeasible?
- 2 Describe progressive alignment and its characteristic failure.
- 3 State the circularity and how it is handled.
- 4 Why are the alignment and the tree not independent?
- 5 Name five tree methods and what each assumes.
- 6 What is the useful question about an assumption?
- 7 What does a bootstrap value measure?
- 8 What does it not measure, and when do they diverge?
- 9 What does the bootstrap exclude, and why does that matter?
- 10 What does an honest tree figure contain?